MatrixAlgebra

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2026-09-21

```
# load libraries
library(Matrix)
```

```
## Warning: package 'Matrix' was built under R version 4.4.3
```

```
library(haven)
```

```
A = matrix(c(2,4,3,3,-2,5), nrow=2, ncol=3,byrow = TRUE)
A
```

```
##      [,1] [,2] [,3]
## [1,]    2    4    3
## [2,]    3   -2    5
```

```
B = matrix(c(1,4,2,1,3,4), nrow=3, ncol=2,byrow = TRUE)
B
```

```
##      [,1] [,2]
## [1,]    1    4
## [2,]    2    1
## [3,]    3    4
```

```
C = matrix(c(1,3,5,6,2,1), nrow=2, ncol=3,byrow = TRUE)
C
```

```
##      [,1] [,2] [,3]
## [1,]    1    3    5
## [2,]    6    2    1
```

```
D = matrix(c(4,2,2,6), nrow=2, ncol=2,byrow = TRUE)
D
```

```
##      [,1] [,2]
## [1,]    4    2
## [2,]    2    6
```

```r
E = matrix(c(1,-1,2,-1,3,1,2,1,10), nrow=3, ncol=3,byrow = TRUE)
E
```

```
##      [,1] [,2] [,3]
## [1,]    1   -1    2
## [2,]   -1    3    1
## [3,]    2    1   10
```

```r
X  = matrix(c(1,2,3,1,4,6,5,7), nrow=4, ncol=2,byrow = TRUE)
X
```

```
##      [,1] [,2]
## [1,]    1    2
## [2,]    3    1
## [3,]    4    6
## [4,]    5    7
```

```r
u = matrix(c(1,3), nrow=1, ncol=2,byrow = TRUE)
u
```

```
##      [,1] [,2]
## [1,]    1    3
```

```r
v = matrix(c(2,7), nrow=2, ncol=1,byrow = TRUE)
v
```

```
##      [,1]
## [1,]    2
## [2,]    7
```

```r
qa = A+C
qa
```

```
##      [,1] [,2] [,3]
## [1,]    3    7    8
## [2,]    9    0    6
```

```r
#qb = A+B # non conformable - cannot do
qc = A %*% B
qc
```

```
##      [,1] [,2]
## [1,]   19   24
## [2,]   14   30
```

```r
#qd = A %*% C # non conformable - cannot do
qe = u %*% D %*% t(u)
qe
```

```
##      [,1]
## [1,]   70
```

```r
qf = u %*% v
qf
```

```
##      [,1]
## [1,]   23
```

```r
qg = t(A+C)
qg
```

```
##      [,1] [,2]
## [1,]    3    9
## [2,]    7    0
## [3,]    8    6
```

```r
qh = 3 * C
qh
```

```
##      [,1] [,2] [,3]
## [1,]    3    9   15
## [2,]   18    6    3
```

```r
qi = det(D)
qi
```

```
## [1] 20
```

```r
qj = solve(D)
qj
```

```
##      [,1] [,2]
## [1,]  0.3 -0.1
## [2,] -0.1  0.2
```

```r
qk = det(E)
qk
```

```
## [1] 3
```

```r
ql = solve(E)
ql
```

```
##           [,1] [,2]       [,3]
## [1,]  9.666667    4 -2.3333333
## [2,]  4.000000    2 -1.0000000
## [3,] -2.333333   -1  0.6666667
```

```r
qm = u %*% solve(D) %*% t(u)
qm
```

```
##      [,1]
## [1,]  1.5
```

```r
qn = B %*% A
qn
```

```
##      [,1] [,2] [,3]
## [1,]   14   -4   23
## [2,]    7    6   11
## [3,]   18    4   29
```

```r
qo = X %*% t(X)
qo
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    5    5   16   19
## [2,]    5   10   18   22
## [3,]   16   18   52   62
## [4,]   19   22   62   74
```

```r
A2 = matrix(c(9,2,1,2,4,5,1,5,3), nrow=3, ncol=3,byrow = TRUE)
A2
```

```
##      [,1] [,2] [,3]
## [1,]    9    2    1
## [2,]    2    4    5
## [3,]    1    5    3
```

```r
question2 = det(A2)
question2
```

```
## [1] -113
```

```r
B2 = matrix(c(6,8,7,6,8,9,2,3,7,2,5,2,6,3,2,1), nrow=4, ncol=4,byrow = TRUE)
B2
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    6    8    7    6
## [2,]    8    9    2    3
## [3,]    7    2    5    2
## [4,]    6    3    2    1
```

```r
question3 = solve(B2)
question3
```

```
##      [,1]  [,2] [,3]  [,4]
## [1,]  0.8 -1.50 -2.0  3.70
## [2,] -1.5  2.75  3.5 -6.25
## [3,] -2.0  3.50  5.0 -8.50
## [4,]  3.7 -6.25 -8.5 14.55
```

```
A3 = matrix(c(2,3,3,6), nrow=2, ncol=2,byrow = TRUE)
A3
```

```
##      [,1] [,2]
## [1,]    2    3
## [2,]    3    6
```

```
B3 = matrix(c(1,0,0,1), nrow=2, ncol=2,byrow = TRUE)
B3
```

```
##      [,1] [,2]
## [1,]    1    0
## [2,]    0    1
```

```
q5a = A3 %*% B3
q5a
```

```
##      [,1] [,2]
## [1,]    2    3
## [2,]    3    6
```

```
q5b = B3 %*% A3
q5b
```

```
##      [,1] [,2]
## [1,]    2    3
## [2,]    3    6
```

```
# We get the same matrix!
# This matrix is called the identity matrix because it acts like a 1 in matrix algebra.
# If you multiply a matrix by an identity matrix, you will get that original matrix.
```

```
S = matrix(c(4,3,1,3,9,1,1,2,1), nrow=3, ncol=3,byrow = TRUE)
S
```

```
##      [,1] [,2] [,3]
## [1,]    4    3    1
## [2,]    3    9    1
## [3,]    1    2    1
```

```
q6 = det(S)
q6
```

```
## [1] 19
```

```r
# The determinant of a covariance matrix like the one we have here is "generalized variance".
# Its telling us how much variance there is overall on average.
```

```r
dataforQ8 = read_sav("MatrixAlgebra.sav")
regression <- lm(y ~ x1 + x2, data = dataforQ8)
summary(regression)
```

```
##
## Call:
## lm(formula = y ~ x1 + x2, data = dataforQ8)
##
## Residuals:
##       1       2       3       4       5
##  1.0512 -0.3819 -1.1457 -0.9606  1.4370
## attr(,"format.spss")
## [1] "F8.2"
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.3543     1.8256   1.837    0.208
## x1            0.9921     0.5629   1.762    0.220
## x2           -0.3898     0.5959  -0.654    0.580
##
## Residual standard error: 1.666 on 2 degrees of freedom
## Multiple R-squared:  0.7109, Adjusted R-squared:  0.4218
## F-statistic: 2.459 on 2 and 2 DF,  p-value: 0.2891
```

```r
# We get the same numbers as what we did in excel!
```